

Faculty Vitae: Shafigh Mehraeen

1. Education

B.Sc. Mechanical Engineering, University of Tehran, Iran 2001

M.Sc. Mechanical Engineering, Stanford University, CA, 2006

PhD Mechanical Engineering, Stanford University, CA, 2011

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago,

Assistant Professor, 2016 – 2024, Full-Time

Associate Professor, 2024 – Present, Full-Time

Massachusetts Institute of Technology, MA, Chemistry Research Fellow, 2013-2016, Full-Time

Georgia Institute of Technology, GA, Chemistry Research Associate, 2011-2013, Full-Time

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

2015, Certificate of Teaching, MIT Teaching Program

2014, Certificate of Teaching, Vanderbilt Teaching Program

5. Current membership in professional organizations

American Institute of Chemical Engineering (AIChE)

American Chemical Society (ACS)

6. Honors and awards

2018, 2019, 2023, College of Engineering Teaching Award, UIC,

2014, Ascher Shapiro Award, MIT

7. Service activities

- Session Organizer, AIChE Midwest Regional Conference, Division of Self-Assembly of Nanostructures, Chicago, (2020 – Present)
- Session Organizer, APS, Division of Fluid Dynamics, Chicago (2020)
- Academic Editor, IntechOpen, London, UK (2019 – Present)
- Undergraduate Research Mentor at UIC (2016- on going)
- Graduate Admission Committee Member: Responsible for reviewing applications for M.S. and Ph.D. programs, reviewing instructional units and graduate courses concerning general policies for graduate education at UIC (2016- on going)
- Reviewer: NSF panel reviewer in CBET program

- Journal reviewer: Physical Review Letters, Physical Review E, Physical Review B, Organic Electronics, Physical Chemistry Letters, Physical Chemistry C, Langmuir (2016 - on going)

8. Selected Publications

1. S. Raghavan, B.P. Chaplin, and S. Mehraeen, Elucidating the impact of electrocatalyst surface and applied potential on the decarboxylation of PFOA radical on Ti_4O_7 [112] facet using DFT Simulations, Journal of Physical Chemistry, in press, (2025).
2. Y.S. Khalid, S.M.M. Islam, S. Mehraeen, and B.P. Chaplin, Reactive-transport modeling of oxidation pathways of insensitive high munitions in porous flow-through electrodes, ACS ES&T Engineering, in press, (2025).
3. S. Raghavan, B.P. Chaplin, and S. Mehraeen, Small-molecule adsorption energy predictions for high- throughput screening of electrocatalysts, Journal of Chemical Information and Modeling, 63(17), 5529- 5538 (2023).
4. D. Christiansen, and S. Mehraeen, Prediction of zwitterion hydration and ion association using machine learning, Soft Matter, 19, 3179-3189 (2023).
5. Z. Luo, and S. Mehraeen, Unraveling the impact of template geometry and confinement on template- assisted self-assembly of nanoparticles, Journal of Molecular Systems Design and Engineering, 8(8), 1075-1086 (2023).
6. B. K. Choi, J. Kim, Z. Luo, T. Hyeon, S. Mehraeen, S. Park, and J. Park, Shape Transformation Mechanism of Gold Nanoplates, ACS Nano, 17(3), 2007- 2018 (2022).
7. Z. Luo, and S. Mehraeen, Free energy contributions to the template-assisted self-assembly of nanopar- ticles using steered molecular dynamics simulations: Implications for molecular- specific imaging and sensing, Journal of Applied Nano Materials, 6(1), 283-295 (2022).
8. D. Christiansen, and S. Mehraeen, Hydration and ion interactions of zwitterionic homopolymers with varying carbon spacer lengths, Journal of Chemical Engineering Research and Design, 100, 174-183 (2022).

9. Professional Development

Mentor training workshop, UIC (2025)